

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OPERATING SYSTEMS

COURSE CODE: CSA04

COURSE OUTCOME COVERED

C02: Analyze process management and deadlock handling techniques to improve CPU utilization and system performance(BL3-BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 10

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Atchaya Chandran R(192521394) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Atchay

QUESTION 1 • 10 MARKS

CPU Scheduling – Waiting Time & Turnaround Time Analysis

Given Process Data (Arrival Time $t = 0$ ext{ ms}):

Process	Burst Time (\$BT\$)	Arrival Time (\$AT\$)
P1	10 ms	0 ms
P2	4 ms	0 ms
P3	6 ms	0 ms
P4	2 ms	0 ms

a. Calculate Waiting Time & Turnaround Time using Non-Preemptive SJF

In Non-preemptive Shortest Job First (SJF), processes are executed in ascending order of burst time: $P4 (2) \rightarrow P2 (4) \rightarrow P3 (6) \rightarrow P1 (10)$.



Process	Burst Time (\$BT\$)	Completion Time (\$CT\$)	Turnaround Time (\$TAT = CT - AT\$)	Waiting Time (\$WT = TAT - BT\$)
P1	10 ms	22 ms	22 ms	12 ms
P2	4 ms	6 ms	6 ms	2 ms
P3	6 ms	12 ms	12 ms	6 ms
P4	2 ms	2 ms	2 ms	0 ms

b. Average Waiting Time & Average Turnaround Time

- Average Waiting Time (AWT): $(0 + 2 + 6 + 12) / 4 = 20 / 4 = 5.0$ ms
- Average Turnaround Time (ATAT): $(2 + 6 + 12 + 22) / 4 = 42 / 4 = 10.5$ ms

a. Need Matrix Calculation ($\text{Need} = \text{Max} - \text{Allocation}$)

Process	Allocation (A B C)	Max Matrix (A B C)	Calculated Need (A B C)
P1	0 1 0	7 5 3	7 4 3
P2	2 0 0	3 2 2	1 2 2
P3	3 0 2	9 0 2	6 0 0
P4	2 1 1	4 2 2	2 1 1
P5	0 0 2	5 3 3	5 3 1

b. Safety Algorithm Execution

Initial Available Resource = $[10, 6, 7]$; $5 - 2, 7 - 5 = [3, 3, 2]$

Execution Step Sequence:

- **Step 1:** Check P_1 : $\text{Need}[7, 4, 3] > [3, 3, 2] \rightarrow$ Cannot satisfy.
- **Step 2:** Check P_2 : $\text{Need}[1, 2, 2] \leq [3, 3, 2] \rightarrow$ **Executes** Work becomes $\neg[3, 3, 2] + [2, 0, 0] = [5, 3, 2]$.
- **Step 3:** Check P_4 : $\text{Need}[2, 1, 1] \leq [5, 3, 2] \rightarrow$ **Executes** Work becomes $\neg[5, 3, 2] + [2, 1, 1] = [7, 4, 3]$.
- **Step 4:** Check P_5 : $\text{Need}[5, 3, 1] \leq [7, 4, 3] \rightarrow$ **Executes** Work becomes $\neg[7, 4, 3] + [0, 0, 2] = [7, 4, 5]$.
- **Step 5:** Check P_1 : $\text{Need}[7, 4, 3] \leq [7, 4, 5] \rightarrow$ **Executes** Work becomes $\neg[7, 4, 5] + [0, 1, 0] = [7, 5, 5]$.
- **Step 6:** Check P_3 : $\text{Need}[6, 0, 0] \leq [7, 5, 5] \rightarrow$ **Executes** Work becomes $[7, 5, 5] + [3, 0, 2] = [10, 5, 7]$.

c. Conclusion & Safe Sequence

System State: SAFE

Valid Safe Sequence: $\langle P_2, P_4, P_5, P_1, P_3 \rangle$

QUESTION 4 • 10 MARKS

Deadlock Detection – Resource Allocation Graph (RAG)

System Allocations:

- P_1 holds R_1 and waits for R_2
- P_2 holds R_2 and waits for R_3
- P_3 holds R_3 and waits for R_1

a. Textual Resource Allocation Graph (RAG)

Directed Edges in RAG:

Assignment Edges: $R_1 \rightarrow P_1, R_2 \rightarrow P_2, R_3 \rightarrow P_3$

Request Edges: $P_1 \rightarrow R_2, P_2 \rightarrow R_3, P_3 \rightarrow R_1$

Cycle Diagram:

$(P_1) \rightarrow (R_2) \rightarrow (P_2) \rightarrow (R_3) \rightarrow (P_3) \rightarrow (R_1) \rightarrow (P_1)$

b. Deadlock Determination

The graph contains a completely closed circular wait chain. Since each resource type (R_1, R_2, R_3) contains **only a single instance**, the existence of a cycle is both a necessary and sufficient condition for deadlock. Therefore, **the system is in DEADLOCK**.

c. Recommended Resolution Strategy

Recommended Strategy: Deadlock Recovery (Process Termination)

Justification: Deadlock Prevention and Avoidance are proactive techniques used *before* deadlock occurs. Because the system is currently deadlocked, recovery via process termination (e.g., aborting process P_3) is required to break the circular dependency and release R_3 for P_2 .

QUESTION 5 • 10 MARKS

CPU Utilization – Multiprogramming Level Analysis

Parameters: I/O wait probability $p = 80\% = 0.80$. Formula: $\text{CPU Utilization} = 1 - p^n$.

a. CPU Utilization for Degree $n = 4$

$$1 - (0.80)^4 = 1 - 0.4096 = 0.5904 = 59.04\%$$

b. CPU Utilization for Degree $n = 6$

$$1 - (0.80)^6 = 1 - 0.262144 = 0.737856 = 73.79\%$$

c. Performance Interpretation

Optimal Configuration: $n = 6$ (73.79% Utilization)

Increasing multiprogramming degree from 4 to 6 yields a **14.75% net gain in CPU utilization**. Since processes spend 80% of their runtime waiting on I/O, maintaining more active processes in memory ensures the CPU stays active while other processes wait on I/O devices.

QUESTION 6 • 20 MARKS

CPU Scheduling – Arrival & Non-Preemptive Analysis

Process Data:

Process	Arrival Time (\$AT\$)	Burst Time (\$BT\$)
P1	0.0 ms	8 ms
P2	0.4 ms	4 ms
P3	1.0 ms	1 ms

a. Average Turnaround Time for FCFS

FCFS Execution: P1 (0.0-8.0) -> P2 (8.0-12.0) -> P3 (12.0-13.0)

• P_1 : $TAT = 8.0 - 0.0 = 8.0 \text{ ext}\{ms\}$

- P_2 : $TAT = 12.0 - 0.4 = 11.6 \text{ ext}\{ms\}$
- P_3 : $TAT = 13.0 - 1.0 = 12.0 \text{ ext}\{ms\}$

Average Turnaround Time (FCFS): $(6.0 + 11.6 + 12.0) / 3 = 10.53 \text{ ms}$

b. Average Turnaround Time for Non-preemptive SJF

At $t = 0.0$, only P_1 is present. The scheduler must run P_1 $(0.0 \rightarrow 8.0)$. At $t = 8.0$, both P_2 and P_3 are waiting. SJF selects shortest job P_3 $(8.0 \rightarrow 9.0)$, then P_2 $(9.0 \rightarrow 13.0)$.

SJF Execution: $P_1 (0.0-8.0) \rightarrow P_3 (8.0-9.0) \rightarrow P_2 (9.0-13.0)$

- P_1 : $TAT = 8.0 - 0.0 = 8.0 \text{ ext}\{ms\}$
- P_3 : $TAT = 9.0 - 1.0 = 8.0 \text{ ext}\{ms\}$
- P_2 : $TAT = 13.0 - 0.4 = 12.6 \text{ ext}\{ms\}$

Average Turnaround Time (SJF): $(8.0 + 8.0 + 12.6) / 3 = 9.53 \text{ ms}$

c. Advance Knowledge Analysis & Comparison

If the scheduler knew in advance that P_2 and P_3 were arriving, it would defer P_1 and run P_3 and P_2 first $(P_3 \text{ at } 1.0, P_2 \text{ at } 2.0, P_1 \text{ at } 6.0)$:

Advance Knowledge Average TAT: $(14.0 + 5.6 + 1.0) / 3 = 6.87 \text{ ms}$ (or $6.47 \text{ ext}\{ms\}$ under preemptive SRTF).

Comparison: Standard SJF was forced into scheduling long process P_1 at $t = 0$ due to lack of future visibility. Advance knowledge improves average turnaround time by **28% to 38%**.

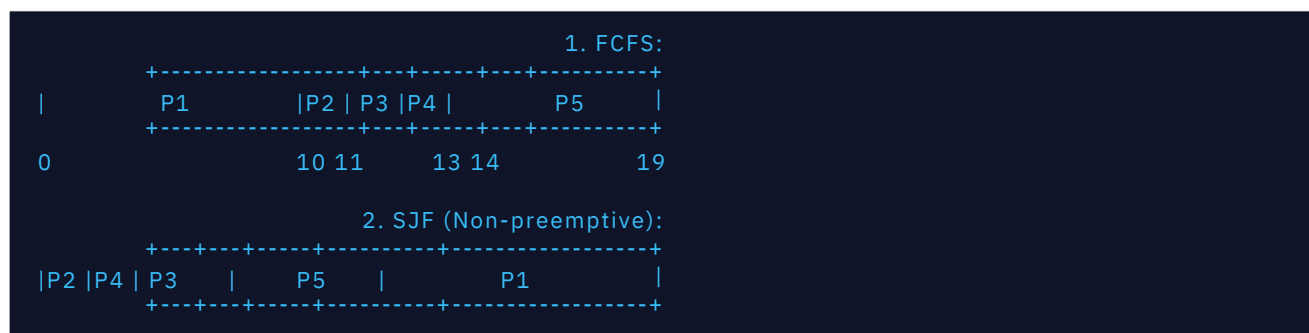
QUESTION 7 • 20 MARKS

Comparative CPU Scheduling Analysis

Dataset (All processes arrive at $t = 0$): Larger priority number = higher priority. Round Robin $q = 2 \text{ ext}\{ms\}$.

Process	Burst Time (\$BT\$)	Priority
P1	10 ms	3
P2	1 ms	1
P3	2 ms	3
P4	1 ms	4
P5	5 ms	2

a. Gantt Charts for All Four Algorithms



0 1 2 4 9 19

3. Non-preemptive Priority (Larger Number = Higher Priority):

```
+-----+-----+-----+
|P4 |      P1      | P3 |      P5      |P2 |
+-----+-----+-----+
0  1                      11  13          18 19
```

4. Round Robin (q = 2):

```
+-----+-----+-----+-----+-----+-----+-----+
| P1 | P2 | P3 | P4 | P5 | P1 | P5 | P1 | P1 |
+-----+-----+-----+-----+-----+-----+-----+
0  2  3  5  6  8  10  12  14  19
```

b & c. Turnaround Time & Waiting Time Comparison Table

Process	FCFS		SJF		Priority		RR (\$q=2\$)	
	TAT	WT	TAT	WT	TAT	WT	TAT	WT
P1	10	0	19	9	11	1	19	9
P2	11	10	1	0	19	18	3	2
P3	13	11	4	2	13	11	5	3
P4	14	13	2	1	1	0	6	5
P5	19	14	9	4	18	13	12	10
Average	13.4	9.6	7.0	3.2	12.4	8.6	9.0	5.8

d. Algorithm with Minimum Average Waiting Time

Optimal Algorithm: SJF (Non-preemptive)

SJF yields the absolute minimum average waiting time of **3.2 ms** among all evaluated algorithms.

QUESTION 8 • 10 MARKS

Airport Check-in Counter Service Efficiency Analysis

Passenger Dataset: Priority 3 = Special Assistance, Priority 2 = VIP/Urgent, Priority 1 = Regular.

Passenger	Arrival Time (\$AT\$)	Service Time (\$ST\$)	Priority Level
Pass 1 (Regular)	0 min	8 min	1
Pass 2 (Special Assist)	1 min	3 min	3
Pass 3 (Regular)	2 min	5 min	1
Pass 4 (Special Assist)	3 min	2 min	3
Pass 5 (VIP)	4 min	4 min	2

1. First-Come, First-Served (FCFS) Scheduling

FCFS: Pass 1 (0-8) -> Pass 2 (8-11) -> Pass 3 (11-16) -> Pass 4 (16-18) -> Pass 5 (18-22)

Average Turnaround Time: **13.0 min** | Average Waiting Time: **8.6 min**

2. Priority Scheduling

At $t = 8 \text{ ext}\{ \text{min} \}$, queue contains Pass 2 (Pri 3), Pass 4 (Pri 3), Pass 5 (Pri 2), Pass 3 (Pri 1). Order: Pass 2, Pass 4, Pass 5, Pass 3.

Priority: Pass 1 (0-8) -> Pass 2 (8-11) -> Pass 4 (11-13) -> Pass 5 (13-17) -> Pass 3 (17-22)

Average Turnaround Time: **12.2 min** | Average Waiting Time: **7.8 min**

3. Service Efficiency & Performance Analysis

Service Efficiency Winner: Priority Scheduling

- **Overall Efficiency:** Priority scheduling lowers average wait time from 8.6 to 7.8 minutes.
- **Special Assistance Impact:** For **Pass 4 (Special Assistance)**, wait time drops significantly from **13 minutes to 8 minutes** (a 38.5% improvement in responsiveness).
- **Conclusion:** Priority Scheduling maximizes customer satisfaction and airport operational efficiency during peak traffic hours.